Inequalities: Rational Functions

Questions 1–22

Question 1.

a) Sketch the graph of

$$y = \frac{x-4}{x}$$

b) Find the values of x, in set notation, where:
   i) $y > 0$
   ii) $y \leq 0$

c) Find the values of x, in interval notation, where:
   i) $y < 0$
   ii) $y \geq 0$

Question 2. Solve the following inequalities, writing your answers in set notation:

a) $\frac{1}{x} > 0$
b) $\frac{1}{x-3} < 0$
c) $\frac{2}{x+1} \geq 0$
d) $\frac{5}{2x+3} \leq 0$
e) $\frac{8}{4-7x} > 0$

Question 3. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{1}{x} > 2$
b) $\frac{1}{x-2} < 3$
c) $\frac{3}{x+4} \geq -5$
d) $\frac{4}{3-2x} \leq 7$
e) $\frac{10}{6-3x} > 9$

Question 4. Francesca is attempting to solve the following inequality:

$$\frac{5}{x-3} \geq 9 - x$$

Her working is shown below:

$$\begin{aligned} &\frac{5}{x-3} \geq 9 - x \\ \Rightarrow\ &5 \geq (9-x)(x-3) \\ \Rightarrow\ &5 \geq -27 + 12x - x^2 \\ \Rightarrow\ &x^2 - 12x + 32 \geq 0 \\ \Rightarrow\ &(x-4)(x-8) \geq 0 \\ \Rightarrow\ &x \in (-\infty, 4) \cup (8, \infty) \end{aligned}$$

a) Identify the error Francesca has made.
b) Write out a correct solution for his problem, giving your answer in interval notation.

Question 5. Solve the following inequalities, writing your answers in set notation:

a) $\frac{x+1}{x} > 0$
b) $\frac{x-1}{2x} < 0$
c) $\frac{2x-3}{4x} \geq 0$
d) $\frac{5-3x}{6x} \leq 0$

Question 6. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{x+2}{x} > 1$
b) $\frac{x-4}{x} < 1$
c) $\frac{2x+1}{2x} \geq 2$
d) $\frac{5-3x}{x} \leq -4$

Question 7. Solve the following inequalities, writing your answers in set notation:

a) $\frac{x}{x+1} > 0$

b) $\frac{2x}{x-3} < 0$

c) $\frac{4x}{2x+3} \geq 0$

d) $\frac{3x}{3-x} \leq 0$

Question 8. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{x-3}{x+1} > 0$

b) $\frac{x+4}{x-5} < 0$

c) $\frac{2x+1}{x+3} \geq 0$

d) $\frac{5-x}{2x+5} \leq 0$

Question 9. Solve the following inequalities, writing your answers in set notation:

a) $\frac{x+3}{x-2} > 1$

b) $\frac{x-6}{x+1} < 1$

c) $\frac{x+4}{2x-1} \geq -3$

d) $\frac{7-x}{x-8} \leq 5$

Question 10. Solve the following inequalities, writing your answers in set notation:

a) $0 < \frac{2x+5}{x-5} < 1$

b) $-1 < \frac{3x+1}{2-x} < 1$

c) $-2 \leq \frac{5-x}{4x+1} \leq 2$

Question 11. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{2}{x} < 2x - 3$
b) $\frac{5}{2x+1} \leq 3x - 2$
c) $\frac{x+3}{x} > x - 1$
d) $\frac{x-5}{2x} < 2 - x$
e) $\frac{x}{x-4} \geq 2x + 1$
f) $\frac{3x}{5-2x} \leq 2x - 3$
g) $\frac{x+2}{x+6} > x - 5$
h) $\frac{4-x}{6-5x} \geq 2x - 1$
i) $\frac{3-2x}{2-3x} \leq 6 - x$

Question 12. Solve the following inequalities, writing your answers in set notation:

a) $\frac{(x-1)(x-2)}{x} > 0$
b) $\frac{(x+3)(x-5)}{x} < 0$
c) $\frac{(2x-1)(3x+4)}{2x} \leq 0$
d) $\frac{(3-x)(1+2x)}{4x} \geq 0$

Question 13. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{x-3}{(x+1)(x+2)} > 0$

b) $\frac{x+4}{(x-2)(x-5)} < 0$

c) $\frac{2x+1}{(x+3)(x-4)} \geq 0$

d) $\frac{3-5x}{(2x+1)(x+3)} \leq 0$

Question 14. Solve the following inequalities, writing your answers in set notation:

a) $\frac{(x-4)(x-3)}{x+2} > 0$

b) $\frac{(x+1)(x+5)}{x-3} < 0$

c) $\frac{(2x-3)(x+2)}{x-1} \geq 0$

d) $\frac{(4-x)(x-6)}{2x+3} \leq 0$

Question 15. It is given that

$$\frac{x-3}{x+5} > x+12$$

a) Show that

$$\frac{(x+9)(x+k)}{x+5} < 0$$

where k is an integer to be found.

b) Hence solve the inequality, writing your answer in interval notation.

Question 16. Solve the following inequalities, writing your answers in set notation:

a) $\frac{(3x-1)(x+1)}{x+2} \geq 12$

b) $\frac{(8-x)(2x+5)}{3x+1} \leq 4$

c) $\frac{(2-5x)(4-x)}{6-x} < -4$

Question 17. Given that $\frac{x^2-9}{5x-2} \leq 3-x$

a) Show that

$$\frac{(px+q)(x-3)}{5x-2} \leq 0$$

where p and q are integers to be found.

b) Hence solve the inequality, writing your answer in interval notation.

Question 18. Find the values of the constants a, b and c, where $a < b$, given that the inequality

$$\frac{(x-a)(x-b)}{x-c} \geq 0$$

has the following solutions:

a) $x \in [2,6] \cup (8,\infty)$

b) $x \in [-3,2) \cup [5,\infty)$

Question 19. Find the values of the constants a, b and c, where $a > b$, given that the inequality

$$\frac{(a-x)(x-b)}{x-c} \geq 0$$

has the following solutions:

a) $x \in (-\infty, -4) \cup [-2, 4]$

b) $x \in (-\infty, 5] \cup (6, 7]$

Question 20. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{(x+1)(x+2)}{(x+3)(x+4)} < 0$

b) $\frac{(x+2)(4-x)}{(x-1)(x+3)} \geq 0$

c) $\frac{(x-3)(5-x)}{(2-x)(6-x)} \leq 0$

Question 21. By solving a relevant inequality, show that the solution to the equation $e^x + 2 = \frac{x+5}{2x-2}$ lies in the interval $1 < x < 3$

Question 22. Given that k is a constant where $k > 5$, solve the following inequalities, writing your answers in set notation:

a) $\frac{2x}{x-k} > x + k$

b) $\frac{x-4}{k-x} < 4 - x$

c) $\frac{x^2-k^2}{x+3} \geq k - x$